

ANN (Artificial Neural Network)

❑ What is ANN?

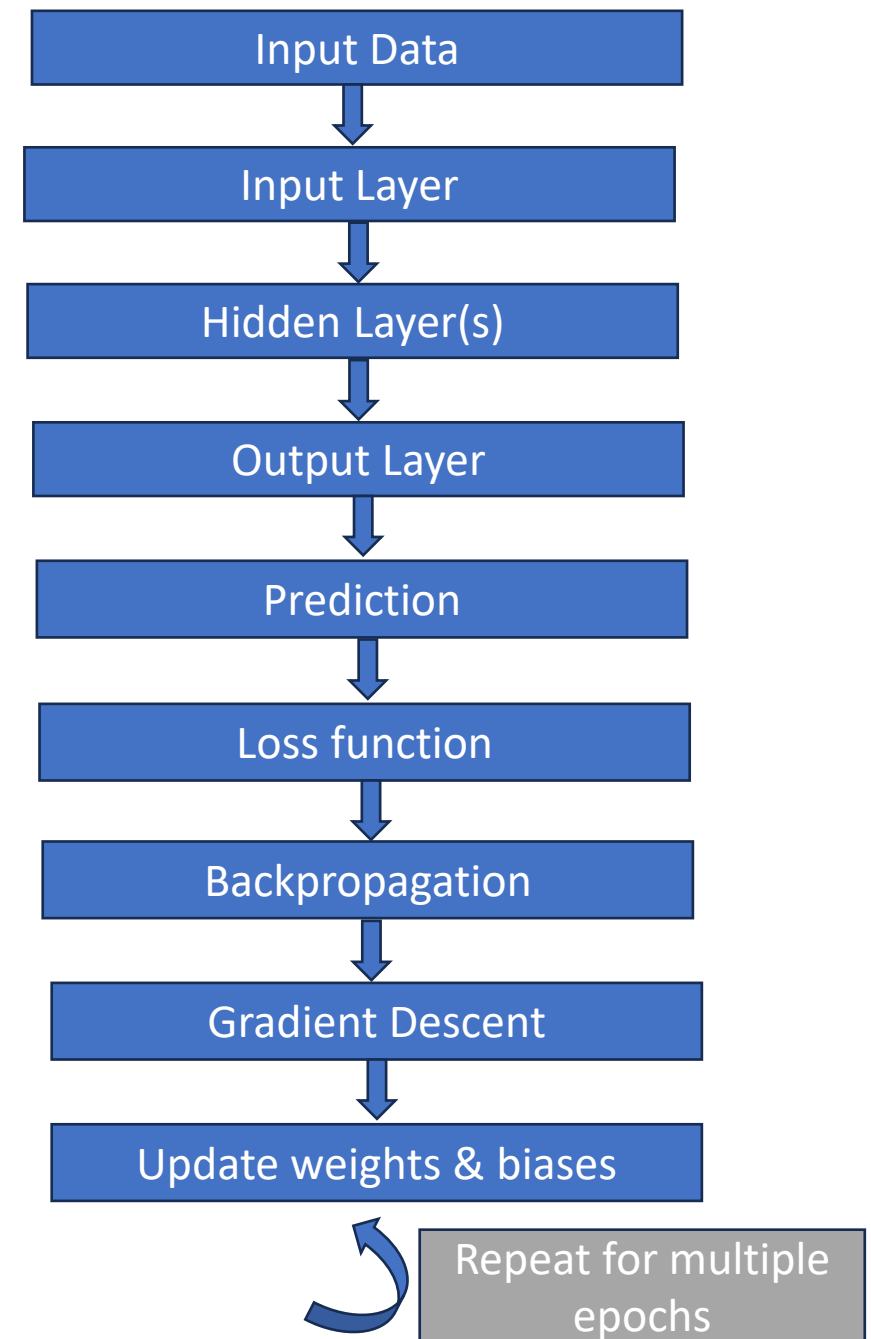
- Inspired by the structure of the human brain.
- Consists of interconnected artificial neurons.
- Learns patterns from training data.

❑ Basic Structure of ANN

- Input Layer
- Hidden Layer(s)
- Output Layer

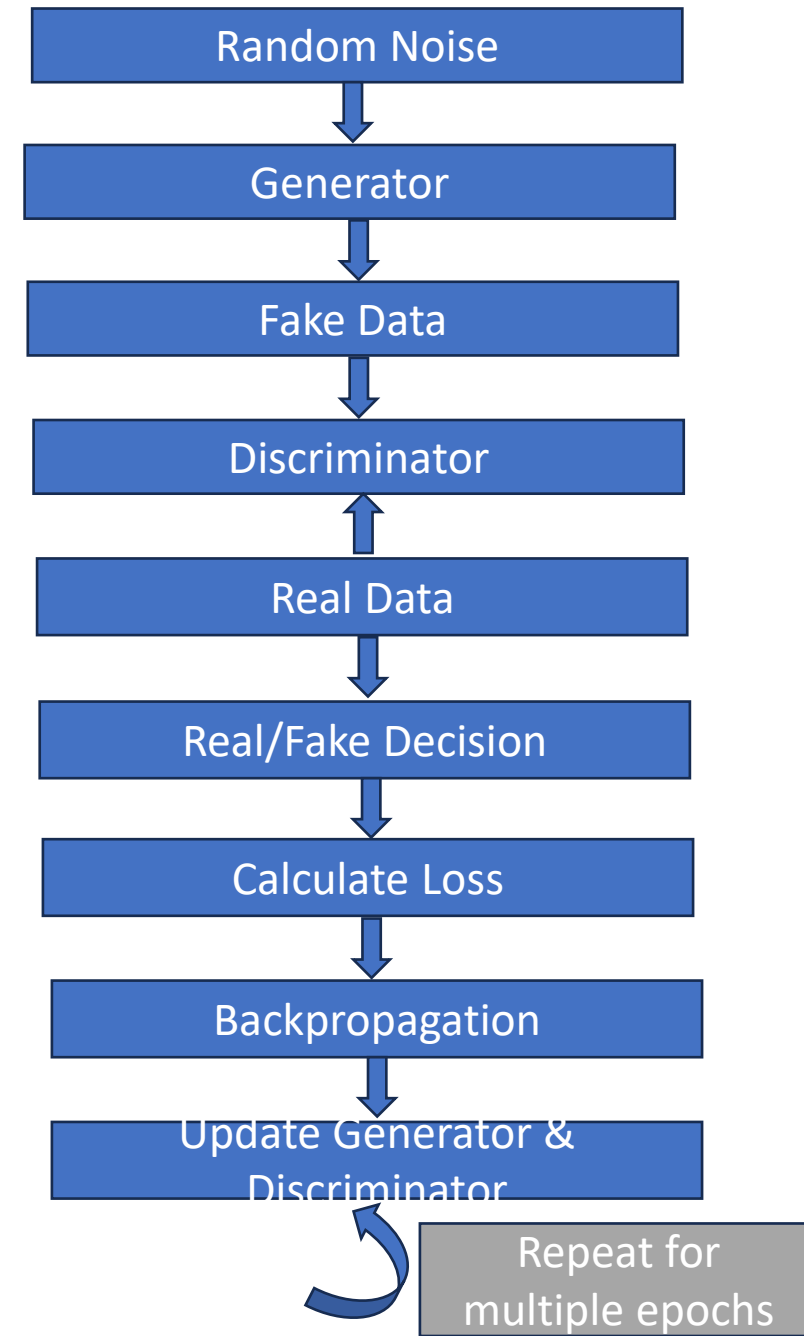
Training Process

- Initialize weights and biases randomly.
- Perform Forward Propagation to generate predictions.
- Compute the Loss.
- Apply Backpropagation to calculate gradients.
- Update parameters using Gradient Descent.
- Repeat for multiple epochs until the loss is minimized.



GAN

- Generative Adversarial Network
- Consists of two neural networks :
 - Generator
 - Discriminator
- Both networks are trained adversarially



Code implementation and result

- Implemented a GAN using Pytorch
- CIFAR-10 dataset
- Built a Generator & Discriminator network
- Trained both networks alternately for 10 epochs
- Generated 16 synthetic images after training.
- Adversarial learning observed through Generator and Discriminator losses.

